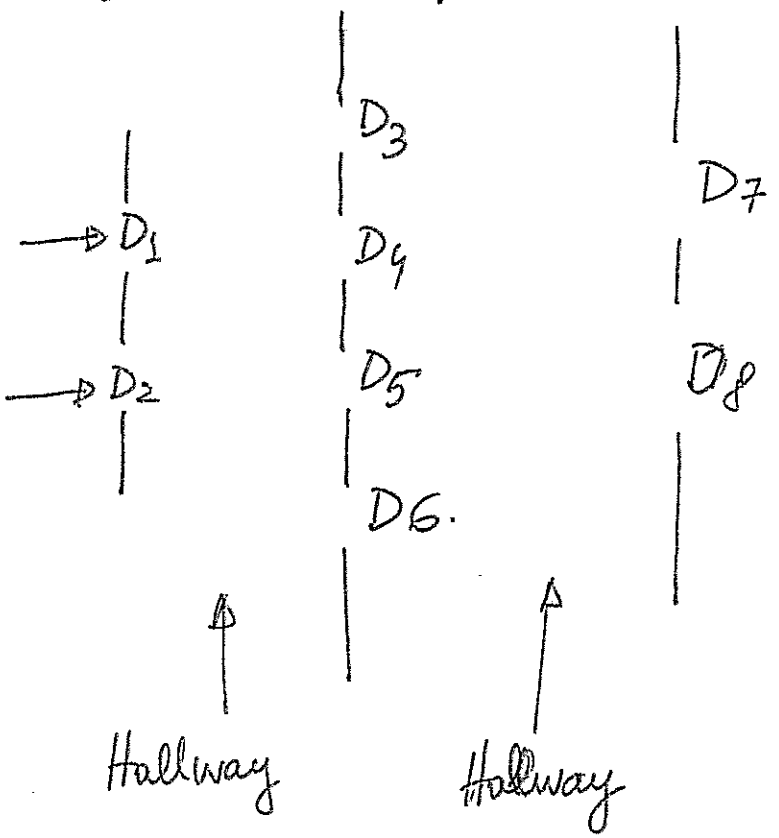
Lecture 2 Counting methods

The basic principle of counting

- Suppose that we perform 2 experiments.
- Experiment 1 has m possible outcomes.
- For each outcome of experiment 1, there are n possible outcomes of experiment 2.
- Then, together, there are $m \times n$ possible outcomes of the two experiments.



- 2×4 ← Basic principle of counting
- $2 \times 4 \times 2$ ← Generalized principle of counting

Example How many different 7-place license plates are possible if the first 3 places are to be occupied by letters and the final 4 by numbers?

Solution Assume we use an alphabet with 26 letters. There are 10 1-digit numbers. Therefore, the generalized principle of counting gives

$$26 \times 26 \times 26 \times 10 \times 10 \times 10 \times 10$$
$$= 175,760,000.$$

Permutations

Definition Let A be a set containing n elements. Any ordered set, of the same size as A, that can be formed with the elements of A is called a permutation of A

Example $A = \{a, b, c\}$.

Ordered sets:

$\{a, b, c\}$; $\{a, c, b\}$; $\{b, a, c\}$; $\{b, c, a\}$; $\{c, a, b\}$; $\{c, b, a\}$.

- $3 \cdot 2 \cdot 1 = 3! = 6$ ways of permuting A.

Result The number of ways in which we can permute n different items is $n!$

Example The number of different ways we can arrange 6 books on the shelf = the number of ways in which we can permute 6 books = $6!$

Arrangements

$$A_{k,n} = \frac{n!}{(n-k)!}$$

Definition I. Let A be a set containing n elements. An ordered subset of k elements of A is called an arrangement of A taken k at a time.

NOTE The definition above is a standard math definition. Your textbook (Definition 1.7.1) calls this "permutation of n taken k at a time". Both terms are fine. In what follows we give an equivalent definition to I (Def. 1.7.1 in your textbook).

Definition I'

- Let A be a set with n elements.
- Consider the following experiment:

(*)
- Select 1 element of A, and do not put it back in A.
- Select a 2nd element, from the remaining set and, again, don't put it back.
- ⋮
- Select a k-th element from the remaining set.

The process described in (*) can be rephrased as:

" Select k elements of A, one at a time, <u>without replacement</u>".

- Consider, as the resulting outcome, the subset of k elements, in the <u>order</u> they were selected.

Each such outcome is called an arrangement (permutation) of n elements taken k at a time.

Result 1 The number of permutations of n token k at a time is

$$P_{n,k} =: n \cdot (n-1) \cdot \ldots \cdot (n-k+1).$$

- Notice that:

$$P_{n,k} = n(n-1)\cdots(n-k+1) \cdot \frac{(n-k)\cdot(n-k-1)\cdots 1}{(n-k)\cdot(n-k-1)\cdots 1}$$

$$= \frac{n!}{(n-k)!}$$

- Use when ranking (order) matters!

Example How many committees of 3 people can be selected from a group of 20?

A possible Answer: $20 \cdot 19 \cdot 18 = \frac{20!}{(20-3)!} = \frac{20!}{17!}$

NOTE The solution above gives

{Jon, Jane, Jill},
{Jane, Jon, Jill},
{Jill, Jon, Jane},
.....

We need to refine it!

NOTE (Sampling with replacement).

If the process ⊛ at page 25 were replaced by

" Select k elements of, one at a time, each time putting the selected element back in A ",

then the number of ways in which we could get an ordered subset of k elements is:

$$\underbrace{n \cdot n \cdot \ldots \cdot n}_{k \text{ times}} = n^k .$$

Combinations

- The number of <u>different</u> groups of size r that can be selected from a set of n objects (when the order of selection is not important) is denoted by $\binom{n}{r}$ and called "combination of n taken r at a time" or "n chooses r", also denoted by $C_{n,r}$.

- <u>Formula</u>: $\binom{n}{r} = \frac{n!}{r!(n-r)!}$ for $0 \le r \le n$.

- <u>Convention</u>: $0! = 1$; $\binom{n}{0} = \frac{n!}{n!0!} = 1$; $\binom{n}{n} = 1$.

- <u>Example</u>: How many <u>different</u> committees of 3 people can be selected from a group of 20?

<u>Answer</u>: $\binom{20}{3} = \frac{20!}{17!3!} = \frac{\cancel{17!} \cdot 18 \cdot 19 \cdot 20}{\cancel{17!} \cdot 1 \cdot 2 \cdot 3} = 1140$.

Example (combinations):

How many DIFFERENT groups of 3 could be selected from the 5 items A, B, C, D, E?

(DIFFERENT = DIFFERENT LETTERS)

- The generalized basic principle of counting (or "arrangements"):

$5 \times 4 \times 3$ ways of selecting 3 letters out of 5, when the order in which the letters appear is relevant.

- Permutations:

Each group of 3 letters appears $3!$ times.

(ABC, BAC, CAB, ACB, BCA, CBA).

Our reasoning:

$$\frac{5 \times 4 \times 3}{3!} = \frac{n(n-1)\dots(n-r+1)}{r!} = \frac{n!}{(n-r)!\,r!}$$

for $n = 5$

$r = 3$.

The binomial theorem

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k},$$

for all $x, y \in \mathbb{R}$ and any $n \geq 0$.

Example How many possible subsets are there of a set consisting of n elements?

Solution:

$\binom{n}{0} = 1$ subsets with 0 elements

$\binom{n}{1} = n$ subsets with 1 element

. . . . . .

$\binom{n}{k}$ subsets with k elements

. . . . . .

$\binom{n}{n} = 1$ subsets with n elements.

$$\sum_{k=0}^{n} \binom{n}{k} = \sum_{k=0}^{n} \binom{n}{k} \cdot 1^k \cdot 1^{n-k} = (1+1)^n = 2^n,$$

q.e.d.

Result The number of possible divisions of n distinct objects into k distinct groups of respective sizes $n_1, n_2, \ldots, n_k$, with $n_1 + n_2 + \ldots + n_k = n$ is:

$$\binom{n}{n_1, n_2, \ldots, n_k} =: \frac{n!}{n_1! n_2! \cdot \ldots \cdot n_k!},$$

called the multinomial coefficient.

Reading assignment: Th. 1.9.1.

Example I have 10 books. I want to keep 3 in the office, 2 at home and 5 in the car. How many possible divisions of my 10 books into 3 groups are possible:

$$\frac{10!}{2! \cdot 3! \cdot 5!}$$

Computing probabilities in "simple spaces" using counting methods

(Sampling without replacement)

Example 1 A committee of 5 is to be chosen out of a group consisting in 6 men and 9 women. If we select at random, what is

$P(E)$, if

E = "Committee consists of 3 men and 2 women"?

Solution: S = sample space = all possible 5 member-committees.

$|S|$ =: The number of elements in $S = \binom{15}{5}$.

$$|E| = \binom{6}{3} \cdot \binom{9}{2}$$

$\binom{9}{2}$ ← Nr. of ways 2 different women can be chosen out of 9.

$\binom{6}{3}$ ↑ Nr. of ways in which 3 distinct men can be chosen out of 6

↑ Basic principle of counting.

Since we are told that "a committee of 5" is to be chosen at random, we implicitly assume that each combination of 5 out 15 is equally likely. Therefore, our sample space is simple, and we can use the rule at page 19:

$$P(E) = \frac{\binom{6}{3}\binom{9}{2}}{\binom{15}{5}} = \frac{240}{1001}.$$

Example 2 In the game of poker, what is the probability that one is dealt a full house?

(Full house = "3 of a kind" + 1 pair.)

Solution: • S = all possible 5-card hands; assume they are all equally likely.

• E = one is dealt a full house.

$$|S| = \binom{52}{5}.$$

- $\binom{4}{2}\binom{4}{3}$ ways to deal a given pair + a certain three of a kind.

- 13 choices for the kind of pair.
  After this has been chosen, we have 12 choices for the denomination of the "3 of a kind".

Therefore: $|E| = 13 \cdot 12 \cdot \binom{4}{2} \cdot \binom{4}{3}$

and so

$$P(E) = \frac{|E|}{|S|} \approx 0.0014$$

q.e.d.

Further properties: the union of 3 events

Result For any events A_1, A_2 and A_3:

$$P(A_1 \cup A_2 \cup A_3) = P(A_1) + P(A_2) + P(A_3) - [P(A_1 \cap A_2) + P(A_2 \cap A_3) + P(A_1 \cap A_3)] + P(A_1 \cap A_2 \cap A_3).$$

Reading assignment The union of a finite number of events, pages 47-48 book.

Example

- A school is offering 3 language classes:
  - one in Spanish (S)
  - one in French (F)
  - one in German (G).
- The classes are open to any of the 100 students.
- There are:

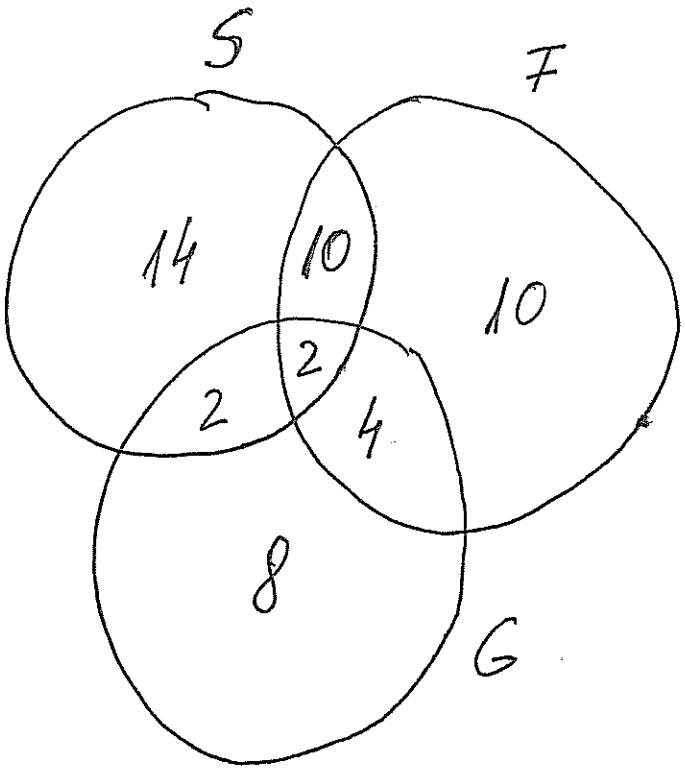
28 students in S
26 " - " in F
16 " - " in G.

12 students in both S and F
4 " - " - " S and G
6 " - " - " F and G
2 " - " in all F, S and G.

(a) If a student is chosen at random, what is the probability that he or she is not in any of these classes?

(b) If a student is chosen randomly, what's the probability that he/she is taking exactly one class?

Solution



(a) $P(\text{"not in } \underline{\text{any}} \text{ class"}) = P[(\text{"In } \underline{\text{some}} \text{ class"})^c]$

$= 1 - P[\text{"In one of the classes"}]$

$= 1 - P(S \cup F \cup G)$

$= 1 - \frac{28+26+16-12-4-6+2}{100} = \frac{1}{2}$

↑ Formula page 34 above

(b) $P(\text{"Exactly one class"}) = \frac{14+10+8}{100}$.

↑ Diagram